

BIOMIMETIC PROSTHETICS REPORT

S.ID: CLI:24066

REQUIREMENTS

Name: kumar

Medical Condition: Traumatic Injury

Prosthetics Type: Tendon & Ligament Repair (Hand)

BIO-ANALYSIS

Material Composition: Polyethylene

Age of Application Years: 3

Usage Frequency Times/Day: 2

Impact Resistance: 7

Wear Resistance: 6

Corrosion Resistance: 7

Temperature Resistance: 8

Integrity: 7

Weight: [0.2, 0.2, 0.2 ,0.2]

Characteristics: [7, 6, 7, 8]

Integrity Level (Prediction): 5.7583

ACTUBIO

Actuation Mechanism: McKibben Actuators (pneumatic muscle)

Biocompatibility Factors: Material selection (elastomers)

Typical Force Output N: 20 - 200 (moderate)

Specific Surface Area: nan

Control Response Time: 10 - 50 (fast)

Energy Density: 0.5 - 1.0 (moderate)

Weight Actubio: 1,000 - 2,000